
Adaptive Residual-Update Steering for Low-Overhead Hallucination Mitigation in Large Vision Language Models

Zhengtao Zou¹ Ya Gao¹ Jiarui Guan¹ Bin Li^{2*} Pekka Marttinen^{1*}

Abstract

Large Vision-Language Models (LVLMs) typically process visual inputs as a prefix to the language decoder. As the model autoregressively generates text, this initial visual information inevitably undergoes “dilution”, leading the model to over-rely on language priors and hallucinate objects. Existing interventions attempt to correct this by contrasting logits or iteratively refining outputs, but they incur prohibitive latency costs. We propose Residual-Update Directed DEcoding Regulation (RUDDER), a framework that counters visual dilution by creating a persistent visual anchor. We extract a robust evidence direction (CARD) directly from the model’s prefill residual updates, and inject it into the decoding process. This injection is modulated by an adaptive gate, the Beta Gate, which acts as a trust mechanism and ensures the visual reminder is applied only when necessary. Experiments on LLaVA-1.5 (7B/13B), Idefics2, InstructBLIP, and Qwen2.5-VL demonstrate that RUDDER consistently mitigates hallucination (with greedy decoding, RUDDER reduces CHAIR_S by an average of 24.4% and CHAIR_i by 23.6% relative) and scales effectively across architectures, all while maintaining 96.0% throughput. The code is available at RUDDER.

1. Introduction

Large Vision-Language Models (LVLMs) usually encode visual inputs as a prefix to the language decoder. Since the model generates text autoregressively, this initial vi-

sual information inevitably undergoes “dilution” (Liu et al., 2024b), resulting in the model over-relying on internal language priors (Leng et al., 2023). This phenomenon is a primary driver of object hallucination. As illustrated in Figure 1 (Left), the model drifts from the visual context, hallucinating a “cup” based on the linguistic association with “surface,” despite no such object being present.

To mitigate this, existing Inference-Time Intervention (ITI) methods usually employ Contrastive Decoding (Leng et al., 2023) or Iterative Refinement (Zhang et al., 2025). However, as summarized in Figure 1(Right), these approaches face a severe bottleneck: they typically require multiple forward passes (extra FWD) or external classifiers, incurring prohibitive latency costs. This undermines the efficiency of the autoregressive loop, making real-world deployment difficult.

To resolve this efficacy-efficiency trade-off, we propose a direct solution: *Instead of correcting the model after it drifts, we structurally prevent the dilution of visual information.* Our key insight is that the visual signal is maximally preserved in the residual stream during the mandatory prefill stage. By capturing this signal before it fades, we can create a lightweight “visual anchor” without any additional forward pass.

We introduce Residual-Update Directed DEcoding Regulation (RUDDER). RUDDER extracts the visual evidence as a Contextual Activation Residual Direction (CARD) vector from the prefill stage. During decoding, RUDDER continuously “reminds” the LLM of this evidence by injecting the CARD vector into the hidden states. To ensure this injection is context-aware (“Per-token Adaptive” feature in Figure 1), we employ Beta Gate, an adaptive gate based on the Beta Distribution. This gate functions as a trust mechanism: it dynamically reinforces the injection when the generation aligns with the visual anchor, and suppresses it when the model is processing non-visual (syntactic) tokens.

Our contributions are:

1. We propose CARD, a method to extract a persistent visual anchor from prefill residual updates. CARD keeps the model grounded in the visual context throughout

^{*}Equal contribution ¹Aalto University, Espoo, Finland
²Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China. Correspondence to: Bin Li <b.li2@siat.ac.cn>, Pekka Marttinen <pekka.marttinen@aalto.fi>.

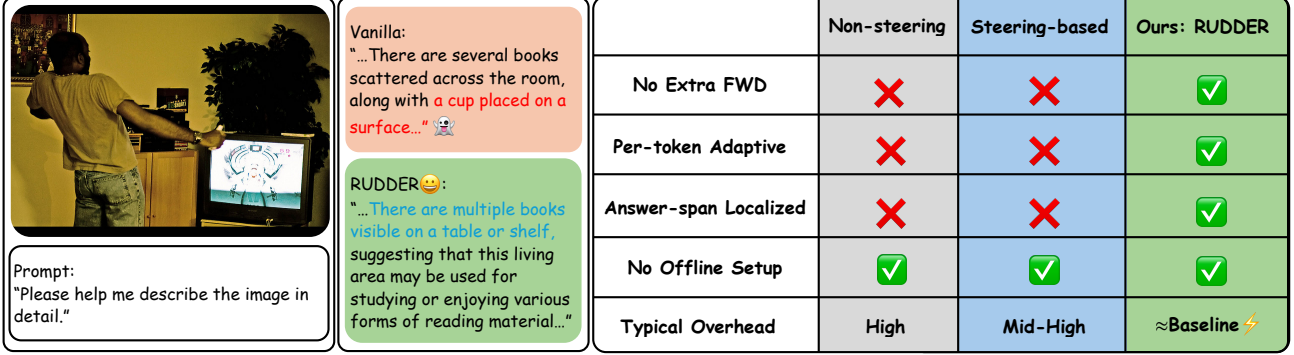
 Prompt: "Please help me describe the image in detail."	Vanilla:	Non-steering	Steering-based	Ours: RUDDER
	"...There are several books scattered across the room, along with a cup placed on a surface..."	✗	✗	✓
	RUDDER:	✗	✗	✓
	"...There are multiple books visible on a table or shelf, suggesting that this living area may be used for studying or enjoying various forms of reading material..."	✗	✗	✓
		✓	✓	✓
		High	Mid-High	≈Baseline

Figure 1. (Left) An example where the vanilla LLaVA-1.5-7B (Liu et al., 2024a) hallucinates objects. Erroneous text is marked in red, while RUDDER’s corrected, factual output is in blue. (Right) A comparison showing that unlike existing non-steering and steering-based methods, RUDDER provides adaptive, low-overhead control without requiring extra forward passes.

the decoding process.

2. We design the Beta Gate, a training-free adaptive gating mechanism based on the Beta Distribution. It ensures that the visual reminder is applied precisely, balancing grounding with fluency.
3. We demonstrate that RUDDER is general and works with different base models. It scales to larger models, such as LLaVA-1.5-13B, and generalizes to modern architectures with distinct fusion strategies, such as Qwen2.5-VL and InstructBLIP.
4. Our method introduces little overhead to the generation process. RUDDER operates within a single forward pass with negligible latency ($< 4\%$), significantly outperforming multi-pass methods.

Conflict of Interest Disclosure. The authors declare no financial conflicts of interest related to this work.

2. Related Work

Our research is situated at the intersection of inference-time intervention (ITI) and probabilistic gating.

Inference-time intervention. ITI aims to guide a model’s generative behavior without modifying its weights. We group existing methods based on where they act on the computation path.

Non-steering methods operate at the output logits or adjust the decoding strategy. Many of these methods recalibrate final logits to improve visual grounding, but often at the cost of significant latency due to extra forward passes. For instance, VCD (Leng et al., 2023) uses perturbed images to create a negative context, PAI (Liu et al., 2024b) subtracts unconditional (text-only) logits, and MARINE (Zhao et al., 2025) employs a classifier-free guidance style. Recent advancements include HALC (Chen et al.,

2024), which introduces adaptive focal-contrast decoding, and AGLA (An et al., 2025), which mitigates hallucinations by assembling global and local attention features. Additionally, DeGF (Zhang et al., 2025) proposes self-correcting decoding via generative feedback, though such iterative refinement typically incurs higher computational overhead. Similarly, DoLa (Chuang et al., 2023) contrasts deep vs. shallow logits to suppress generic text. More efficient alternatives, such as constrained decoding (Hokamp & Liu, 2017) or post hoc editing (Manakul et al., 2023), are typically less adaptive.

Steering methods directly modify the hidden representations to guide the generation trajectory. Most of these methods also incur high computational costs on-the-fly. For example, ASD (Su et al., 2025) steers away from a predefined hallucination direction, and VISTA (Li et al., 2025) injects a signal vector computed from activation differences. VTI (Liu et al., 2025) attempts to mitigate this cost by shifting the computational burden to an offline precomputation step.

Bayesian and Probabilistic Gating. Our work is also inspired by Bayesian and probabilistic gating for uncertainty modeling. This includes concepts from Evidential Deep Learning (Sensoy et al., 2018), which frames outputs as parameters of a Dirichlet distribution for uncertainty quantification. Other relevant work explores stochastic gates. For instance, (Yamada et al., 2020) use stochastic gates based on a relaxation of the Bernoulli distribution for feature selection. More directly related to our method, Beta-LSTM (Song et al., 2019) replaces standard sigmoid gates with ones derived from a Beta distribution, validating the use of Bayesian principles in gating mechanisms.

3. Our Method

In autoregressive LVLMs, visual information tends to fade as the generation sequence lengthens, a phenomenon often referred to as visual fading or attention decay (Liu et al.,

2024b). However, during the initial prefill stage, the fusion of visual and textual information is at its peak. The residual update from the self-attention sublayer, therefore, encodes the net effect of the visual context on the representation of each text token. We hypothesize that by aggregating these updates over the image tokens and text prompt tokens in the prefill span, we can obtain a robust, per-sample vector that captures the direction of visual evidence for the specific input. Our empirical analysis supports this: the extracted CARD vector creates a systematic, image-conditioned rotation away from a text-only (language prior) direction, and this rotation aligns coherently with the downstream steering mechanism. This confirms the aggregated updates provide a meaningful directional signal rather than random noise (a detailed visualization and quantification is in Appendix A.4).

To mitigate object hallucination without the high computational costs of existing steering methods, we present Residual-Update Directed DEcoding Regulation (RUDDER). RUDDER is a low-overhead framework that adaptively steers LVLMs toward visually-grounded generation by injecting a dynamically derived visual evidence vector. Crucially, it delivers context-specific steering within a single forward pass and introduces negligible latency. RUDDER is training-free; its few deployment hyperparameters are selected once on a small held-out validation set, rather than learned from task-specific supervision.

3.1. Preliminaries

The decoder in a Transformer-based LVLM operates on a residual stream, where each sublayer’s output (e.g., self-attention of the decoder layer) is added back to its input. This output, termed the residual updates Δ^l , represents the new information contributed at layer l . We leverage these updates during the two-stage auto-regressive generation process: **1. Prefill Stage:** The model processes the prefill span, comprising both image tokens and text prompt tokens, in a single parallel forward pass to populate a Key-Value cache. During this mandatory step, we extract the CARD vector by aggregating the self-attention residual updates across all tokens in the prefill span. **2. Decoding Stage:** The model generates the output sequentially, one token at a time. It’s during this phase that we employ Beta Gate for adaptive steering.

Scope of the intervention. RUDDER acts in the hidden residual stream rather than at the final logit distribution. Its objective is therefore not to provide a formal guarantee on the token-level likelihood, but to preserve and reassert an input-specific visual evidence direction during decoding. This design matches the failure mode we target: hallucination often arises when the generation trajectory drifts from the visual prefix toward language priors, while the original visual evidence is still present in the model’s internal

computation during prefill.

3.2. CARD Vector: A Zero-Cost Per-Sample Evidence Direction

Motivation. LVLMs fuse visual and textual information through self-attention. The residual update from the self-attention sublayer, therefore, encodes the net effect of the visual context on the representation of each text token. We hypothesize that by aggregating these updates over the image tokens and text prompt tokens in the prefill span, we can obtain a robust, per-sample vector that captures the direction of visual evidence for the specific input (Liu et al., 2024a). Our empirical analysis supports this: the extracted CARD vector creates a systematic, image-conditioned rotation away from a text-only (language prior) direction, and this rotation aligns coherently with the downstream steering mechanism. This confirms the aggregated updates provide a meaningful directional signal rather than random noise (a detailed visualization and quantification is in Appendix A.4).

To identify the optimal layer for extracting the CARD vector, we analyze internal dynamics of LLaVA-1.5-7B (Liu et al., 2024a). We find that intervening in the late decoder layers has the greatest potential to influence the model’s final output. Full analysis is provided in Appendix B.1, Figure 9a, 9c.

Extraction. In a single standard prefill pass with the image and text prompt, we place a lightweight, read-only hook at the *target* decoder layer l and cache the self-attention output for each token i in the prefill span $\mathcal{T}_{\text{pre}}$, denoted $\mathbf{A}_i^l$. In a pre-norm decoder, the residual update is simply the attention output,

$$\Delta_i^l = \mathbf{A}_i^l, \quad (1)$$

We then pool these updates and apply L_2 normalization to obtain a per-sample direction:

$$\mathbf{v}_{\text{CARD}} = \frac{\text{Pool}(\{\Delta_i^l\}_{i \in \mathcal{T}_{\text{pre}}})}{\|\text{Pool}(\{\Delta_i^l\}_{i \in \mathcal{T}_{\text{pre}}})\|_2}, \quad (2)$$

where $\text{Pool}(\cdot)$ can be mean or $\|\Delta_i^l\|$ -weighted mean.

Intuition. Although pooling aggregates strictly over the prefill span, it naturally prioritizes semantic content. Since self-attention updates are magnitude-heavy on informative tokens and lighter on functional syntax, the weighted pooling effectively filters out noise, distilling a robust, input-specific evidence direction that summarizes the net visual influence. This entire process occurs within the single prefill pass and introduces negligible overhead, as no additional forward pass or calibration is required.

3.3. Beta Gate: Adaptive Gating Based on the Beta Distribution

Motivation: A Trust Mechanism. Standard steering often applies a fixed vector, which can be harmful when the visual context is irrelevant (e.g., on syntactic tokens). To address this, we introduce an adaptive gate derived from a probabilistic perspective. Let $\mathbf{h}_{l,t}$ be the hidden state for generating token t at the target layer l . We measure its alignment with the visual evidence via cosine similarity: $s_t = \cos(\mathbf{h}_{l,t}, \mathbf{v}_{\text{CARD}})$.

We use the function of a Beta-Bernoulli conjugate posterior to design the gate. We view s_t as providing pseudo-counts for a latent “visual groundedness” probability. Crucially, this acts as a trust mechanism rather than an error detector:

- **High alignment** ($s_t \uparrow$): Indicates the generation is consistent with visual evidence (high trust). The gate increases to reinforce this valid trajectory.
- **Low/Negative alignment** ($s_t \downarrow$): Indicates uncertainty or non-visual context. The gate suppresses intervention to prevent over-steering and preserve fluency.

This monotone design is intentional. A high value of s_t means the current hidden state is already compatible with the extracted visual evidence, so reinforcing $\mathbf{v}_{\text{CARD}}$ is unlikely to disrupt generation. Conversely, a low or negative value of s_t suggests either non-visual function-token generation or an unstable trajectory, where aggressive steering can hurt fluency or object recall. The gate therefore controls *trust in the current visual alignment*, rather than measuring deviation magnitude.

Gate Formulation. Using this intuition, we calculate the gate parameters using a smooth mapping from s_t :

$$\begin{aligned} \alpha_t &= \text{softplus}(k s_t + c), \\ \beta_t &= \text{softplus}(-k s_t + c), \\ g_t &= \frac{\alpha_t}{\alpha_t + \beta_t}. \end{aligned} \quad (3)$$

Here, k is a sensitivity hyperparameter controlling the steepness of the response, while c is a concentration parameter typically fixed to a robust default (e.g., $c = 1$). We optimize these few parameters on a small held-out validation set (details in Sec. 4.1). To ensure stability, we clamp the gate’s output to a predefined range, $g_t \in [g_{\min}, g_{\max}]$. This prevents the gate from completely shutting off ($g_t \rightarrow 0$) or saturating ($g_t \rightarrow 1$) too readily, making the intervention robust to noise.

For generating each token t in the answer, the final steering update $\mathbf{v}_t^{\text{steer}}$ combines the adaptive gate with a global cap

$$\alpha_{\max}: \quad \mathbf{v}_t^{\text{steer}} = \underbrace{(\alpha_{\max} g_t)}_{\text{adaptive strength}} \mathbf{v}_{\text{CARD}}, \quad (4)$$

This vector is injected into the residual stream immediately after the Self-Attention(SA) operation. The updated hidden state, $\mathbf{h}_{l,t}^{\text{new}}$, is thus computed as:

$$\mathbf{h}_{l,t}^{\text{new}} = (\mathbf{h}_{l,t} + \text{SA}(\mathbf{h}_{l,t})) + \mathbf{v}_t^{\text{steer}}. \quad (5)$$

The term $\alpha_{\max} g_t$ represents the adaptive strength of the intervention, ensuring a strong corrective signal is applied only when needed; the injection is restricted to the answer span.

3.4. Integration and Workflow

Our complete method, Residual-Update Directed DEcoding Regulation (RUDDER), integrates the CARD vector and the adaptive Beta Gate to mitigate hallucination by steering LVLMS toward visually grounded outputs. We refer to this primary, adaptive configuration as RUDDER-Beta. To isolate the impact of adaptivity, we also define a static variant, RUDDER-Add, which injects the CARD vector with a constant strength without the gating mechanism.

As detailed in Algorithm 1, RUDDER can be seamlessly integrated into the standard auto-regressive decoding loop. By operating within a single inference pass, it mitigates hallucination with negligible computational overhead, resolving the common trade-off between efficacy and efficiency. The overall workflow of this approach is illustrated in Figure 2.

4. Experiments

In this section, we validate RUDDER, demonstrating its ability to mitigate hallucination effectively with negligible computational overhead. We conduct a series of experiments across diverse LVLMS architectures and benchmarks to assess performance, general capabilities, efficiency, and hyperparameter sensitivity.

4.1. Experimental Setup

Model Architectures. We evaluate RUDDER on three representative LVLMS with distinct visual-textual alignment mechanisms: LLaVA-1.5-7B (Liu et al., 2024a) and Idedics2-8B (Laurençon et al., 2024) (linear projection), and InstructBLIP (Dai et al., 2023) (Q-former). We also include LLaVA-1.5-13B (Liu et al., 2024a) and Qwen2.5-VL-7B (Bai et al., 2025) in our scalability analysis (Sec. 4.6).

Baselines. We compare RUDDER against state-of-the-art inference-time intervention methods. These include logit-based strategies like DoLa (Chuang et al., 2023),

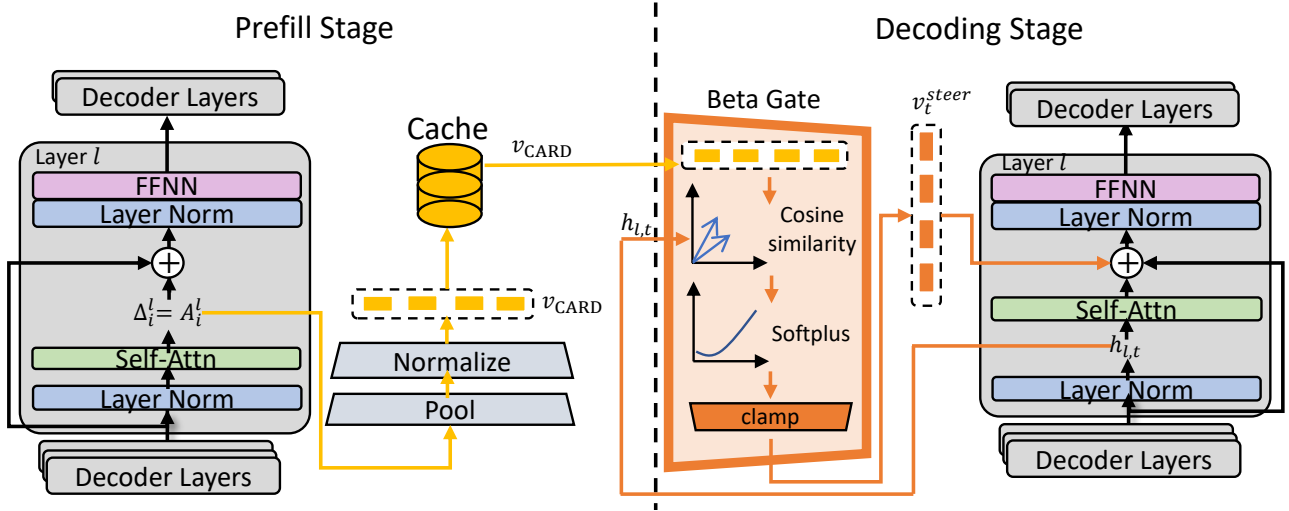


Figure 2. **The overall workflow of RUDDER.** (1) **Prefill Stage** (Yellow): We extract the CARD vector $\mathbf{v}_{\text{CARD}}$ by aggregating residual updates from the target layer. (2) **Decoding Stage** (Orange): At each step t , the Adaptive Beta Gate computes a steering vector $\mathbf{v}_t^{\text{steer}}$ to modulate the residual stream, guiding the model toward visually grounded outputs.

VCD (Leng et al., 2023), PAI (Liu et al., 2024b), and the recent HALC (Chen et al., 2024), AGLA (An et al., 2025), and DeGF (Zhang et al., 2025); as well as steering-based interventions like VISTA (Li et al., 2025). All baseline results were reproduced under identical evaluation settings.

Implementation Details. We configure RUDDER using a lightweight calibration on 100 held-out MSCOCO images, disjoint from the evaluation subset. This calibration is used only to select deployment hyperparameters and is not a training procedure. The process involves two main steps:

- **Step 1: Layer Selection (L).** We perform a linear sweep to maximize CARD vector alignment. We select mid-to-late layers for LLaVA-1.5 ($L = 30$) and Idefics2 ($L = 28$), and an early layer for InstructBLIP ($L = 1$).
- **Step 2: Hyperparameter Tuning.** We grid-search the steering strength α_{max} and gate sensitivity k to minimize CHAIR scores while maintaining $\geq 95\%$ baseline recall. Other parameters are fixed to robust defaults ($c = 1, g \in [0.05, 1]$).

Final settings (α_{max}, k) are (20, 5.0) for LLaVA-1.5, (8.0, 5.0) for Idefics2, and (6.5, 8.0) for InstructBLIP. This one-time setup incurs negligible overhead. We deliberately constrain the search to preserve at least 95% of the vanilla recall, since hallucination reduction is only useful when the model does not lose its general ability to mention correct objects.

Evaluation Benchmarks. We evaluate RUDDER using both specialized hallucination benchmarks and a general capability benchmark. (1) **CHAIR** (Rohrbach et al., 2019): We evaluate open-ended captioning on

MSCOCO (Lin et al., 2015) validation set, reporting sentence-level (CHAIR_S) and object-level (CHAIR_I) metrics. For the CHAIR benchmark, we evaluate 500 samples using the prompt “Please help me describe the image in detail” with a maximum length of 512 tokens. (2) **POPE** (Li et al., 2023): We assess object probing via targeted yes/no questions, reporting Accuracy and F1 scores averaged across random, popular, and adversarial splits. For the POPE benchmark, we use the prompt “Is there a <object> in the image?” and let the models to answer with “yes” or “no”. (3) **MME** (Fu et al., 2024): We use MME to confirm that our method maintains general multimodal capabilities (e.g., reasoning, counting). Following established protocols, we set the temperature to 1.0 and test greedy, beam search (beam=5), and nucleus sampling ($p = 0.9$).

4.2. Results on Hallucination Benchmarks

4.2.1. CHAIR: OPEN-ENDED CAPTIONING

On the CHAIR benchmark, which evaluates hallucination in open-ended captioning, RUDDER demonstrates a strong ability to reduce factual errors while preserving caption quality.

A key challenge in hallucination mitigation is the trade-off with recall: aggressive steering can artificially lower hallucination scores by producing overly simplistic captions. To ensure a fair and practical evaluation, we constrain our analysis to configurations that maintain at least 95% of the vanilla model’s recall (i.e., $\text{Recall}_{\{\text{evaluated methods}\}} \geq 0.95 \times \text{Recall}_{\{\text{vanilla model}\}}$).

Under this constraint, RUDDER-Beta consistently outper-

forms the vanilla baseline across all tested LVLMS and decoding strategies, as shown in Table 1. It achieves average relative reductions of **33.2%** in sentence-level (CHAIR_S) and **28.6%** in object-level (CHAIR_I) hallucination across these three models.

Compared to strong baselines like VCD and DoLa, our method is consistently superior on both metrics. Furthermore, RUDDER-Beta performs on par with the state-of-the-art VISTA and, on average, yields a greater reduction in object-level hallucinations (CHAIR_I).

RUDDER-Beta’s ability to reduce CHAIR_I more effectively than CHAIR_S highlights its precision. We attribute this to the token-wise gating mechanism, which selectively amplifies corrections on visually incongruent or content-noun tokens while leaving already grounded tokens largely unperturbed. This allows RUDDER to preferentially suppress object-level hallucinations without degrading overall caption quality and recall.

4.2.2. POPE: VISUAL QUESTION ANSWERING

Moving from open-ended captioning to a more constrained task, we next evaluate RUDDER on the POPE benchmark for object probing. This benchmark tests the model’s factuality through targeted yes/no questions, offering a different perspective on hallucination. In this setting, RUDDER again demonstrates competitive performance. As shown in Table 2, RUDDER consistently outperforms the vanilla baselines and most competing methods across all tested models. Concretely, RUDDER-Beta improves accuracy by 1.0/0.7/0.5 absolute points (pp) and F1 by 1.6/1.3/0.14 pp on LLaVA-1.5, Idefics2, and InstructBLIP, respectively.

Notably, RUDDER-Beta achieves the highest F1-score and accuracy on both LLaVA-1.5 and Idefics2, surpassing strong steering-based methods like VISTA. While its performance on InstructBLIP is slightly surpassed by VISTA when employing greedy decoding and nucleus sampling, RUDDER remains highly competitive, highlighting its effectiveness as a versatile solution for reducing object hallucination.

4.2.3. ANALYSIS OF ADAPTIVE VS. FIXED-STRENGTH STEERING

A key design choice in RUDDER is the regulation of injection strength. We analyze the trade-off between the adaptive gate (RUDDER-Beta) and the fixed-strength injection (RUDDER-Add). Our experiments show a clear distinction between these variants, guiding the choice based on the task and model architecture.

For complex, open-ended generation (CHAIR), RUDDER-Beta is consistently superior. Its token-wise precision is crucial for suppressing specific hallucinations in long-form text without harming overall recall. In the simpler, binary-

choice POPE task, the distinction is more nuanced. While RUDDER-Beta remains the top performer on LLaVA-1.5 and Idefics2, RUDDER-Add is competitive and even surpasses RUDDER-Beta on InstructBLIP. We hypothesize this is partly because InstructBLIP’s Q-Former provides a highly-condensed visual representation that responds well to a uniform steering signal in a simple setting. For single-token “yes/no” answers, the aggressive push from fixed-strength steering can be sufficient and sometimes more beneficial for certain model architectures.

In summary, RUDDER-Beta is recommended for robust and precise control in complex tasks, while the simpler RUDDER-Add is a powerful option for constrained tasks and certain model architectures.

4.3. Results on Comprehensive Benchmarks

To ensure that hallucination mitigation does not compromise general multimodal capabilities, we evaluate RUDDER on MME benchmark. The results show that RUDDER successfully reduces hallucinations without sacrificing the overall abilities of the tested LVLMS. As demonstrated in Table 3, both RUDDER-Beta and RUDDER-Add achieve higher MME scores than the vanilla models for Idefics2 and InstructBLIP. On LLaVA-1.5, RUDDER’s scores are slightly lower than the vanilla model, but the difference is still acceptable.

4.4. Efficiency Tests

A critical advantage of RUDDER is its low computational overhead, making it practical for real-world deployment. Unlike many state-of-the-art intervention methods that require extra forward passes and significantly increase latency, RUDDER is designed to operate within a single generative pass. We measure the practical latency and throughput of RUDDER against vanilla models and other methods, with results presented in Table 4. All experiments are conducted on a single Nvidia A100 GPU with 80 GB VRAM and a batch size fixed at 1. RUDDER-Beta maintains an average throughput of 96.0% compared to vanilla LVLMS. RUDDER-Add is even more efficient as it bypasses the Beta Gate calculation. In contrast, competing methods that require extra forward passes see a significant drop in efficiency. On average, the throughput of a method like VISTA is only 58.1% of the vanilla models.

4.5. Ablation Studies

We conduct an ablation study on Idefics2 using the CHAIR benchmark to analyze the key hyperparameters: injection layer L , maximum steering strength $\alpha_{\max}$ and the gate sensitivity k .

We first identify the optimal intervention layer, finding Layer

Table 1. Hallucination evaluation on the CHAIR benchmark. We compare RUDDER against state-of-the-art training-free methods with a max generation length of 512 tokens. Note that methods marked with [†] (e.g., DeGF) require iterative updates, incurring significantly higher latency ($> 3\times$) compared to RUDDER’s single-pass efficiency. Best results are **bolded** and second-best are underlined.

Decoding	Method	LLaVA-1.5 (Liu et al., 2024a)		Idefics2 (Laurençon et al., 2024)		InstructBLIP (Dai et al., 2023)	
		CHAIR _S ↓	CHAIR _I ↓	CHAIR _S ↓	CHAIR _I ↓	CHAIR _S ↓	CHAIR _I ↓
Greedy	Vanilla	48.6	13.6	46.6	14.9	39.2	12.8
	DoLa (Chuang et al., 2023)	47.6	13.4	-	-	-	-
	VCD (Leng et al., 2023)	49.8	14.5	-	-	46.4	15.3
	HALC (Chen et al., 2024)	40.3	10.3	-	-	-	-
	AGLA (An et al., 2025)	44.6	12.1	-	-	-	-
	DeGF [†] (Zhang et al., 2025)	34.2	9.2	-	-	-	-
	VISTA (Li et al., 2025)	<u>38.6</u>	11.4	33.5	<u>11.6</u>	<u>27.7</u>	<u>9.7</u>
	RUDDER-Beta (Ours)	39.5	<u>10.5</u>	28.4	10.9	27.1	8.5
	RUDDER-Add (Ours)	42.1	11.8	<u>30.1</u>	11.8	28.3	10.4
Beam Search	Vanilla	52.8	15.6	48.6	14.5	38.2	12.7
	VCD (Leng et al., 2023)	52.4	15.5	-	-	47.4	16.3
	VISTA (Li et al., 2025)	33.9	<u>10.5</u>	32.2	11.8	<u>27.1</u>	<u>9.6</u>
	RUDDER-Beta (Ours)	33.1	9.3	29.2	10.1	26.2	9.5
	RUDDER-Add (Ours)	35.2	10.6	<u>31.4</u>	<u>10.9</u>	27.4	11.1
Nucleus Sampling	Vanilla	55.6	16.0	53.8	16.7	46.0	16.2
	DoLa (Chuang et al., 2023)	49.3	14.8	-	-	-	-
	VCD (Leng et al., 2023)	57.5	17.2	-	-	53.3	19.8
	VISTA (Li et al., 2025)	39.2	<u>11.9</u>	<u>35.5</u>	<u>11.8</u>	<u>29.0</u>	11.3
	RUDDER-Beta (Ours)	<u>39.9</u>	11.0	34.1	11.3	28.9	<u>13.7</u>
	RUDDER-Add (Ours)	41.6	12.1	36.5	12.9	30.1	14.4

Table 2. Performance on the POPE benchmark across three LVLMs. The reported values are the mean accuracy and F1 score, aggregated over the random, popular, and adversarial object splits. The best scores are **bolded**, and the second best scores are underlined.

Decoding	Method	LLaVA-1.5 (Liu et al., 2024a)		Idefics2 (Laurençon et al., 2024)		InstructBLIP (Dai et al., 2023)	
		Acc ↑	F1 ↑	Acc ↑	F1 ↑	Acc ↑	F1 ↑
Greedy	Vanilla	85.34	84.91	78.40	74.86	85.74	84.75
	DoLa (Chuang et al., 2023)	85.51	84.96	-	-	-	-
	VCD (Leng et al., 2023)	85.46	84.87	-	-	85.79	84.89
	PAI (Liu et al., 2024b)	85.98	85.31	-	-	-	-
	VISTA (Li et al., 2025)	<u>86.21</u>	<u>85.42</u>	78.28	74.66	86.25	85.06
	RUDDER-Beta (Ours)	86.53	86.03	78.74	76.52	86.02	84.93
	RUDDER-Add (Ours)	85.92	84.98	<u>78.43</u>	<u>75.91</u>	<u>86.05</u>	<u>85.05</u>
Beam Search	Vanilla	85.46	84.98	78.67	77.55	84.73	84.37
	VCD (Leng et al., 2023)	85.60	85.06	-	-	84.95	84.59
	PAI (Liu et al., 2024b)	85.58	85.01	-	-	-	-
	VISTA (Li et al., 2025)	86.10	85.35	78.40	77.31	85.64	84.61
	RUDDER-Beta (Ours)	86.51	86.19	79.33	77.96	85.54	84.40
	RUDDER-Add (Ours)	85.98	85.02	<u>78.91</u>	<u>77.60</u>	85.71	84.75
Nucleus Sampling	Vanilla	83.00	81.08	74.84	67.78	85.50	84.52
	DoLa (Chuang et al., 2023)	82.94	81.12	-	-	-	-
	VCD (Leng et al., 2023)	82.82	81.90	-	-	85.61	84.65
	PAI (Liu et al., 2024b)	83.17	82.14	-	-	-	-
	VISTA (Li et al., 2025)	83.58	82.21	74.66	67.70	86.12	85.26
	RUDDER-Beta (Ours)	84.02	83.57	75.89	69.69	85.79	84.74
	RUDDER-Add (Ours)	83.20	<u>82.38</u>	<u>74.95</u>	<u>67.84</u>	<u>85.95</u>	<u>84.95</u>

28 is the most effective for the IdeFics2 model, as shown in Figure 3. Focusing on this layer, we then tune the hyperparameters $\alpha_{\max}$ and k . The heatmaps in Figures 4a through 4c reveal a core trade-off: increasing the steering strength ($\alpha_{\max}$) effectively reduces CHAIR scores but at the cost of lower recall. The gate sensitivity k , does not exhibit a simple linear trend; instead, it plays a crucial modulating role in

this trade-off. Ultimately, we find that the best balance for IdeFics2 is achieved with $\alpha_{\max} = 8.0$ and $k = 5.0$. Ablation results for other models are presented in Appendix B.2.

Table 3. Overall performance scores on the MME full evaluation set. Higher scores indicate better general capability across perception, reasoning, and knowledge-based tasks.

Decoding	Method	LLaVA-1.5	Idefics2	InstructBLIP
Greedy	Vanilla	1745.87	1518.84	1566.77
	RUDDER-Beta	1724.17	1540.56	1592.07
	RUDDER-Add	1715.45	1526.03	1585.28
Beam Search	Vanilla	1760.20	1450.59	1539.16
	RUDDER-Beta	1746.66	1484.21	1565.77
	RUDDER-Add	1738.13	1475.80	1560.64
Nucleus Sampling	Vanilla	1752.65	1362.45	1538.18
	RUDDER-Beta	1721.94	1374.77	1556.43
	RUDDER-Add	1713.74	1364.16	1546.01

Table 4. Throughput Comparison (tokens/s). Measurements are conducted using greedy decoding. Higher scores indicate better efficiency.

Method	LLaVA-1.5	Idefics2	InstructBLIP
Vanilla	56.7	47.8	62.3
VCD	30.1	-	-
PAI	29.5	-	-
VISTA	36.1	31.9	28.9
RUDDER-Beta (Ours)	54.9	45.8	59.5
RUDDER-Add (Ours)	55.8	46.5	60.8

4.6. Scalability and Generalization Analysis

To verify that RUDDER is not over-fitted to specific model sizes or architectures, we extend our evaluation to LLaVA-1.5-13B (to verify scalability) and the recent Qwen2.5-VL-7B (to verify architectural generalization).

As shown in Table 5, RUDDER demonstrates robust performance across these diverse settings. On the larger LLaVA-1.5-13B, RUDDER scales effectively, surpassing the VISTA baseline in sentence-level hallucination reduction (CHAIR_S : 39.9 vs. 40.1) and achieving the highest POPE performance (F1: 85.5).

The results on Qwen2.5-VL highlight the precision of our method. While VISTA achieves a lower CHAIR_S , it suffers a regression in POPE performance compared to the vanilla baseline (F1: 87.8 $\rightarrow$ 87.4, Acc: 88.8 $\rightarrow$ 88.6), suggesting that its aggressive steering may compromise the model’s fundamental object recognition capabilities. In contrast, RUDDER achieves the lowest instance-level hallucination rate (CHAIR_I : 7.0) while simultaneously improving POPE performance (F1: 88.1). This confirms that the CARD vector captures a universal and precise visual evidence signal, mitigating hallucinations without the side effect of degrading general recognition accuracy.

4.7. Case Study

Qualitative analysis in Appendix B.3 demonstrates RUDDER’s effectiveness. The case studies show that RUD-

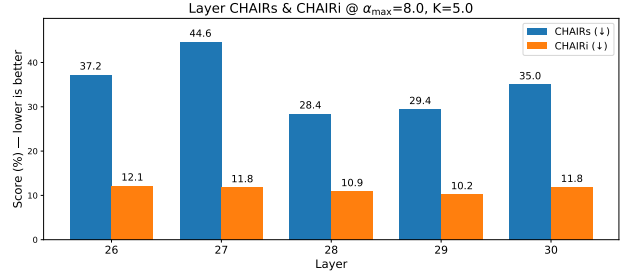


Figure 3. Layer ablation study on Idefics2. We fix ($\alpha_{\max} = 8.0$, $k = 5.0$) and vary the injection layer. Mid-late layers ($L \approx 28-30$) prove most effective, with $L=28$ yielding a strong reduction.

Table 5. Scalability and Generalization Analysis. We assess RUDDER on a larger backbone (LLaVA-1.5-13B) and a distinct modern architecture (Qwen2.5-VL-7B) using Greedy decoding.

Model	Method	CHAIR		POPE	
		$C_S \downarrow$	$C_I \downarrow$	Acc $\uparrow$	F1 $\uparrow$
LLaVA-1.5 (13B)	Vanilla	44.2	11.8	85.7	85.0
	VISTA	40.1	10.6	86.0	85.2
	RUDDER	39.9	10.8	86.3	85.5
Qwen2.5-VL (7B)	Vanilla	35.2	9.5	88.8	87.8
	VISTA	25.1	7.7	88.6	87.4
	RUDDER	26.9	7.0	89.0	88.1

DER not only eliminates object hallucinations present in the vanilla model’s outputs but also produces more conservative content. By avoiding the vanilla model’s confident yet incorrect assertions, RUDDER enhances the model’s overall reliability.

5. Conclusion and Limitations

In this work, we introduce RUDDER, a low-overhead inference-time intervention framework that mitigates LVLMS hallucination using two key innovations: the zero-cost CARD vector, which extracts a per-sample visual evidence from the model’s own residual updates, and the adaptive Beta Gate, which applies a corrective signal with principled, token-wise strength.

Experiments confirm RUDDER achieves state-of-the-art comparable performance on benchmarks like CHAIR and POPE with negligible computational overhead, resolving the common efficacy-efficiency trade-off. RUDDER presents a practical and effective solution for enhancing the reliability of LVLMS in real-world settings.

RUDDER’s primary limitation is its sensitivity to hyperparameters, which need be tuned for each model architecture. Future work could focus on automated hyperparameter optimization to improve its robustness and ease of deployment.

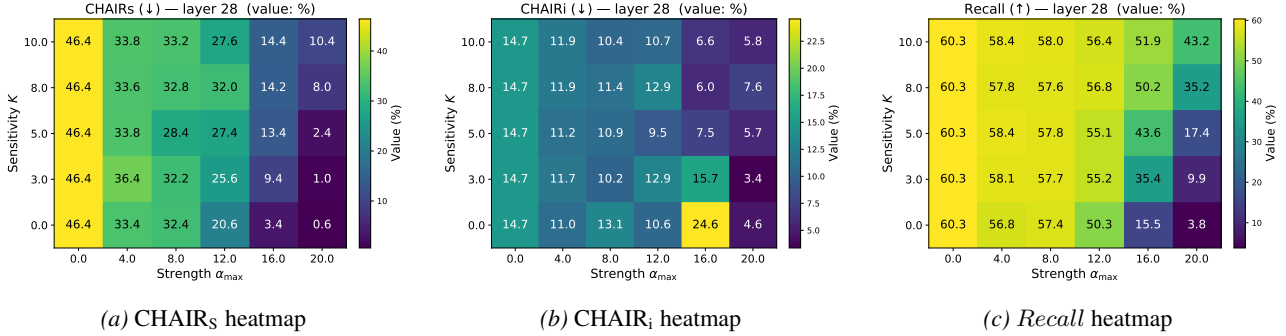


Figure 4. **Hyperparameter sensitivity analysis on Idexics2.** We analyze the trade-off between steering strength $\alpha_{\max}$ and gate sensitivity k at Layer 28. Stronger steering reduces hallucinations (a, b) but may impact recall (c). The optimal balance is found around $\alpha_{\max} = 8.0, k = 5.0$.

Impact Statement

This paper presents RUDDER, a method designed to mitigate object hallucinations in LVLs. Hallucinations pose significant risks in real-world applications, such as autonomous systems, medical imaging analysis, and robotics, where factual errors can lead to safety hazards. By improving the grounding of model outputs in visual evidence, our work contributes to the development of more reliable and trustworthy AI systems.

A core contribution of RUDDER is its computational efficiency. RUDDER operates with negligible overhead. This efficiency aligns with the goals of “Green AI,” reducing the energy footprint of high-quality model inference and lowering the barrier to deploying aligned LVLs in resource-constrained environments.

Acknowledgements

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A. Additional Illustration on the Methodology

A.1. RUDDER Algorithm

Here we present the pseudo-code for RUDDER illustrated in Sec. 3.

A.2. From Naïve Bayes to the Adaptive Gate

Problem setup. At each decoding step t , we want a scalar gate $g_t \in (0, 1)$ that reflects how much the current token should be nudged toward the visual evidence direction $\mathbf{v}_{\text{CARD}}$. Let the alignment statistic be $s_t = \cos(\mathbf{h}_t, \mathbf{v}_{\text{CARD}}) \in [-1, 1]$.

Naïve Bayes view (posterior as a gate). Introduce a latent Bernoulli variable $Z_t \in \{0, 1\}$ indicating whether the token is visually grounded ($Z_t = 1$) or at risk of drifting ($Z_t = 0$). We use the posterior mean $g_t = \mathbb{E}[Z_t | s_t]$ as a *continuous* gate (rather than a hard on/off decision).

Beta-Bernoulli conjugacy with “soft counts”. With a Beta prior $\text{Beta}(\alpha_t, \beta_t)$ on Z_t , the posterior mean is

$$g_t = \frac{\alpha_t}{\alpha_t + \beta_t}.$$

We map the alignment s_t to *positive pseudo-counts* via a smooth, monotone transform:

$$\alpha_t = \text{softplus}(k s_t + c), \quad \beta_t = \text{softplus}(-k s_t + c),$$

where k controls sensitivity and c controls concentration/bias. The softplus ensures strictly positive, numerically stable “counts”.

Algorithm 1 RUDDER (single-pass, test-time steering; fixed target layer ℓ)

```

1: Input: Model  $M$ ; image  $x_{\text{img}}$ , text  $x_{\text{text}}$ ; fixed layer  $\ell$ ; hyperparams  $(\alpha_{\text{max}}, k, c, g_{\text{min}}, g_{\text{max}})$ 
2:  $\mathcal{T}_{\text{pre}} \leftarrow \text{TokenizePrefill}(x_{\text{img}}, x_{\text{text}})$  {image + prompt tokens}
3: Prefill: run  $M$  once (read-only hook at layer  $\ell$ ) to build KV cache and cache  $\{\mathbf{A}_i^\ell\}_{i \in \mathcal{T}_{\text{pre}}}$ 
4:  $\Delta_i^\ell \leftarrow \mathbf{A}_i^\ell$  by Eq. 1 {pre-norm: residual update equals SA output}
5:  $\mathbf{v}_{\text{CARD}} \leftarrow$  by Eq. 2 {Pool  $\rightarrow$   $L_2$ -Normalize over  $\{\Delta_i^\ell\}_{i \in \mathcal{T}_{\text{pre}}}$ }
6: Decode: for  $t = 1, 2, \dots$  {auto-regressive generation}
7:    $s_t \leftarrow \cos(\mathbf{h}_{\ell,t}, \mathbf{v}_{\text{CARD}}^\ell)$ 
8:    $(\alpha_t, \beta_t, g_{\ell,t}) \leftarrow$  by Eq. 3
9:    $g_{\ell,t} \leftarrow \text{clip}(g_{\ell,t}, g_{\text{min}}, g_{\text{max}})$ 
10:   $\mathbf{h}_{\ell,t}^{\text{new}} \leftarrow (\mathbf{h}_{\ell,t} + \text{SA}^{(\ell)}(\mathbf{h}_{\ell,t})) + \mathbf{1}[t \in \mathcal{T}_{\text{ans}}] \cdot \mathbf{v}_t^{\text{steer}}$  {post-SA residual add; answer span only}
11:  emit next token
12: end for
    
```

Properties (useful for calibration). The resulting g_t is monotone in s_t , symmetric $g(-s) = 1 - g(s)$, and bounded in $(0, 1)$. Around $s = 0$, the slope

$$\left. \frac{\partial g}{\partial s} \right|_{s=0} = \frac{k \sigma(c)}{2 \text{softplus}(c)}, \quad \sigma(x) = \frac{1}{1 + e^{-x}},$$

gives a handy knob to set how fast the gate reacts to alignment changes.

Stability: clamping and optional per-token cap. For robustness we clamp $g_t \leftarrow \text{clip}(g_t; g_{\text{min}}, g_{\text{max}})$ to avoid both shutting off ($g_t \rightarrow 0$) and saturating ($g_t \rightarrow 1$). Optionally, we enforce a per-token norm cap τ :

$$\|\alpha_{\text{max}} g_t \hat{\mathbf{v}}_{\text{CARD}}\|_2 \leq \tau, \quad \hat{\mathbf{v}} = \mathbf{v} / \|\mathbf{v}\|_2,$$

which further prevents rare spikes when hidden-state norms vary.

Final update (matches Algorithm 1).

$$\mathbf{v}_t^{\text{steer}} = \underbrace{(\alpha_{\text{max}} g_t)}_{\text{adaptive strength}} \mathbf{v}_{\text{CARD}} \quad (6)$$

$$\mathbf{h}_{\ell,t}^{\text{new}} = (\mathbf{h}_{\ell,t} + \text{SA}(\mathbf{h}_{\ell,t})) + \mathbf{v}_t^{\text{steer}} \quad (7)$$

We apply this only on the answer span using the mask m_t as in Algorithm 1.

Implementation notes.

- We compute s_t with L2-normalized $\mathbf{h}_t$ and $\mathbf{v}_{\text{CARD}}$ (cosine similarity).
- g_t is clamped to $[g_{\text{min}}, g_{\text{max}}]$; the global scale α_{max} controls the maximal push.
- $\mathbf{v}_{\text{CARD}}$ is extracted once during the mandatory prefill pass (zero extra forwards).

Practical calibration recipe. Choose c to set the overall smoothness (typical $c \in [0.5, 1.5]$), then increase k until g_t becomes sufficiently responsive on a small dev set. Finally tune α_{max} and $[g_{\text{min}}, g_{\text{max}}]$ for stability/strength trade-offs.

Complexity. The gate requires only light-weight vector ops during decoding and reuses the prefill to compute $\mathbf{v}_{\text{CARD}}$. Hence no extra forward pass compared to vanilla generation.

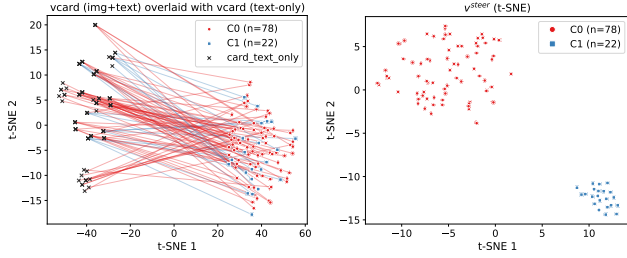
A.3. Clarification

RUDDER maintains the same inference efficiency as the original model, requiring no extra forward passes. The CARD vector is extracted opportunistically during the mandatory prefill pass, and the steering is applied within each step of the subsequent decoding pass.

By no extra forward pass we mean no additional `model.forward` invocations beyond the vanilla prefill and decode; our overhead comes only from cheap per-token vector operations implemented via hooks.

A.4. Visualization and Quantification of the Geometry of $\mathbf{v}_{\text{CARD}}$ and $\mathbf{v}_t^{\text{steer}}$

Setup and link to motivation. Large VLMs fuse vision and text via self-attention; the residual update of this sub-layer thus captures the *net* impact of visual context on token representations. Motivated by this, we aggregate the prefill-phase residual updates to obtain a per-sample direction $\mathbf{v}_{\text{CARD}} \in \mathbb{R}^d$ and define its steering counterpart $\mathbf{v}_t^{\text{steer}} = (\alpha_{\text{max}} \text{gate}) \mathbf{v}_{\text{CARD}}$ (Sec. 3). For each image we export (i) $\mathbf{v}_{\text{CARD}}$ (image+prompt) and its text-only variant, and (ii) $\mathbf{v}_t^{\text{steer}}$. We reduce vectors by PCA ($k=50$) then t-SNE (default perplexity unless noted), and cluster the steering space with KMeans (best silhouette over $K \in \{2, \dots, 10\}$). Figure 5 shows two key views: a *paired* overlay of image+text vs. text-only $\mathbf{v}_{\text{CARD}}$ with one-to-one lines, and the clustered t-SNE of $\mathbf{v}_t^{\text{steer}}$. The overlay reveals systematic sample-wise *rotations* from the language prior (text-only) to



(a) t-SNE overlay of $\mathbf{v}_{\text{CARD}}$ (img+text; colored by $\mathbf{v}_t^{\text{steer}}$ clusters) and text-only $\mathbf{v}_{\text{CARD}}$ (black), with lines linking paired samples. (b) t-SNE of $\mathbf{v}_t^{\text{steer}}$ with KMeans clusters.

Figure 5. Structure in steering space (b) and its sample-wise projection to $\mathbf{v}_{\text{CARD}}$ (a).

the image-conditioned direction, and these rotations point toward coherent steering clusters—visual evidence is therefore *directional* rather than noise, directly supporting our motivation.

Quantifying directional structure. We quantify two effects central to our hypothesis: (i) **Rotation from text-only to image-conditioned CARD:** $\Delta\theta = \arccos(\mathbf{v}_{\text{text}}, \mathbf{v}_{\text{img+txt}})$ shows a tight distribution around $\sim 40^\circ$ (mean $\approx 40.5^\circ$, median $\approx 40.4^\circ$), indicating a consistent, non-trivial visual-induced rotation rather than random drift (Fig. 6a); this effect persists across steering clusters (Fig. 6c). (ii) **Alignment gain w.r.t. steering:** $\langle \mathbf{v}_{\text{img+txt}}, \mathbf{v}^{\text{steer}} \rangle - \langle \mathbf{v}_{\text{text}}, \mathbf{v}^{\text{steer}} \rangle$ is positive on average (mean ≈ 0.239 , median ≈ 0.238) and remains positive across clusters (Figs. 6b,d), showing that image-conditioned $\mathbf{v}_{\text{CARD}}$ is *closer* to the actual steering geometry used by the β -gate. Together, these results substantiate our motivation: aggregating self-attention residual updates yields a robust sample-specific visual-evidence direction that aligns with the downstream steering mechanism.

Notes. Silhouette scores are typically higher for $\mathbf{v}_t^{\text{steer}}$ than for $\mathbf{v}_{\text{CARD}}$, consistent with the gate organizing/scaling directions across samples. We emphasize that t-SNE primarily supports *local* neighborhood interpretation; all scalar statistics are computed in the original vector spaces.

Semantic clustering of CARD. In addition to the paired geometry above, we directly test whether CARD encodes category-level visual semantics. We extract CARD vectors from LLaVA-1.5-7B on four COCO categories with 50 images per category and visualize them with t-SNE. Figure 7 shows that samples from the same visual category form coherent neighborhoods, suggesting that CARD captures image-conditioned semantic structure rather than random activation noise.

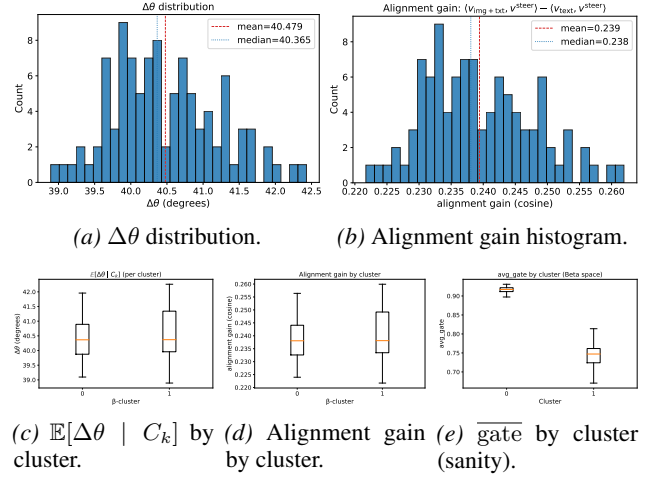


Figure 6. Directional evidence with reflowed layout. (a) Consistent $\mathbf{v}_{\text{text}} \rightarrow \mathbf{v}_{\text{img+txt}}$ rotation; (b) positive alignment gain to $\mathbf{v}^{\text{steer}}$; (c,d) cluster-wise stability; (e) systematic gate differences.

B. Additional Experiment

B.1. LLaVA Internal Dynamic Analysis

As mentioned in Section 3.2, our method is guided by an analysis of the internal dynamics of LLaVA-1.5 (Liu et al., 2024a), with key findings illustrated in Figure 9. By examining the residual update vector from the self-attention module at each layer, we identified two properties that informed our intervention strategy:

Intervention Leverage Peaks in Late Layers. We can find that the magnitude of the residual update, which represents the “leverage” an intervention can have, is not uniform. We found that its strength grows with model depth, peaking in the late decoder blocks (approx. layers 26-32). This indicates that interventions in these layers have the greatest potential to influence the model’s final output (Figure 9a, 9c).

Directional Coherence Collapses in Late Layers. While late layers offer the most leverage, the directional coherence of their update vectors collapses after approximately layer 21 (Figure 9b). Coherence is moderate only in the early-to-mid layers. This suggests that applying a fixed, global steering vector in the high-leverage late layers is suboptimal, as the intervention may be misaligned with the model’s unstable internal state.

B.2. Additional Ablation Results

We provide supplementary ablation results on LLaVA-1.5 and InstructBLIP, as shown in Figure 15 and Figure 16. These analyses complement the main ablation study conducted on Idefics2 in Section 4.4.

We further verify whether the CHAIR-tuned hyperparam-

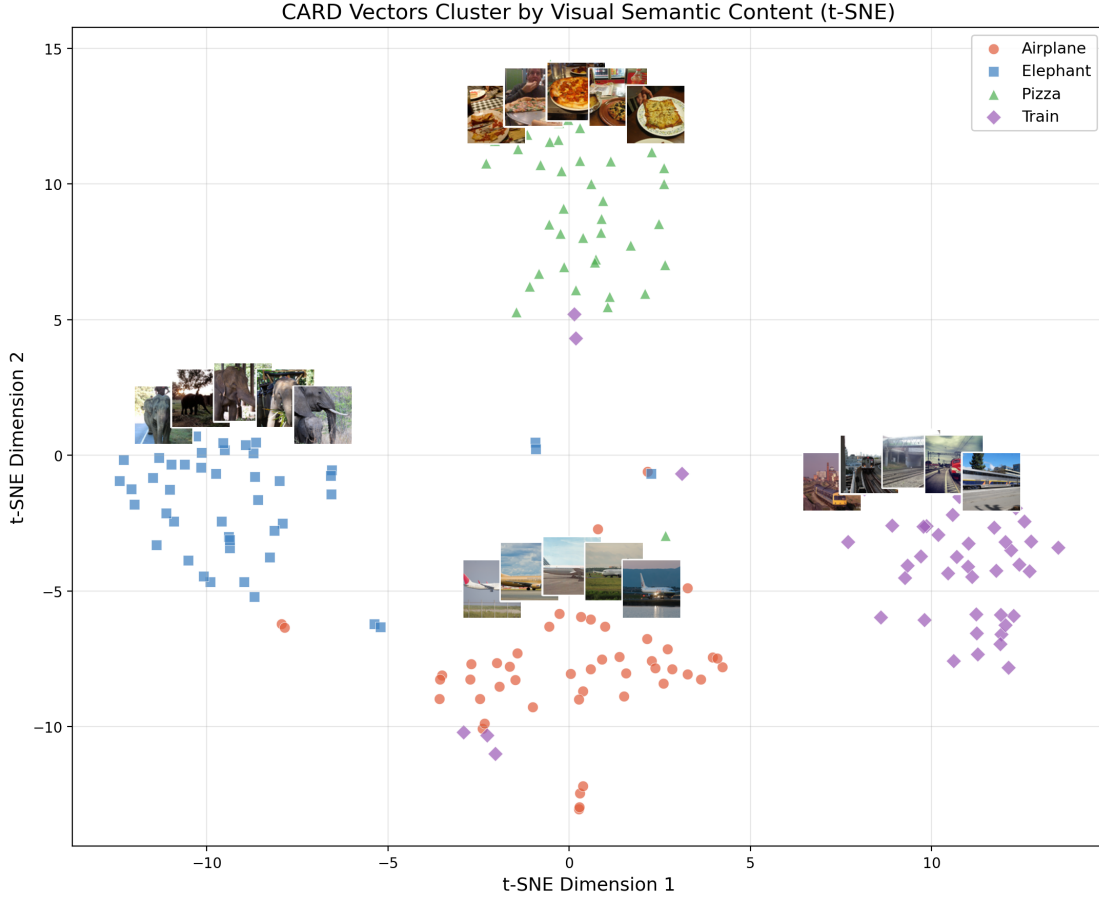


Figure 7. CARD vectors cluster by visual semantic content. We compute CARD vectors from LLaVA-1.5-7B on four COCO categories with 50 images per category and visualize them using t-SNE. The category-level clustering indicates that CARD captures image-conditioned visual semantics rather than random activation noise.

ters transfer to POPE. Figure 8 shows that all tested configurations improve over the vanilla baseline, and the CHAIR-selected setting remains close to the best POPE configuration. This supports the use of a small held-out calibration set rather than benchmark-specific retuning.

The results for both models confirm the same core trends observed with Idefics2. Specifically, the heatmaps reveal a consistent trade-off between hallucination mitigation and recall. As the steering strength α_{max} increases, both CHAIRS and CHAIRI scores improve (decrease), but this is often accompanied by a drop in recall. The gate sensitivity parameter, k , plays a similar, non-linear modulating role in this balance. While the general trade-off is consistent, the optimal hyperparameter values vary by model architecture, highlighting the need for model-specific tuning.

B.3. Case Study

To provide a qualitative illustration of our method’s real-world performance, we present a series of case studies in Figures 12, 13 and 14. From these examples, we can find

that RUDDER is highly effective at eliminating object-level hallucinations. It successfully removes entirely non-existent objects from the captions (e.g., a hallucinated “second person” or “cup”), and corrects misidentified objects (e.g., correctly identifying “skis” instead of “snowboard”). Moreover, RUDDER’s outputs are not only more factually accurate but also more semantically cautious. The corrected descriptions often adopt more conservative language, such as using phrases like “appears to be”, “may be”, or “suggesting that”. By replacing the baseline models’ confident yet incorrect assertions with more grounded and appropriately qualified statements, RUDDER significantly enhances the overall reliability and trustworthiness of the generated text.

To further inspect the behavior of the adaptive gate, we visualize token-level gate values and CARD similarities for two representative samples. Figures 10 and 11 show that visually grounded content tokens generally receive stronger alignment and gate values, whereas tokens associated with unsupported concepts receive lower CARD similarity. These examples complement the caption-level case studies by

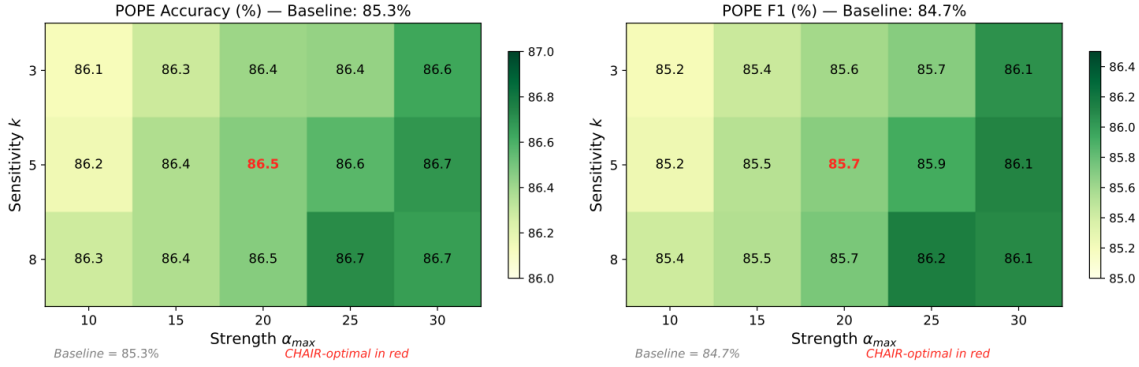


Figure 8. **POPE hyperparameter sensitivity on LLaVA-1.5-7B.** All tested configurations improve over the baseline, with a small spread across the grid. The CHAIR-tuned setting remains close to the best POPE configuration, suggesting that RUDDER is not highly sensitive to the exact choice of $\alpha_{\max}$ and k .

showing that the Beta Gate operates at the intended token granularity.

C. Clarifications Added After Reviewer Discussion

Bayesian-inspired rather than full Bayesian inference.

Our Beta Gate is not intended to perform full posterior inference over model parameters or generated tokens. Instead, it uses the Beta-Bernoulli posterior mean as a structured, bounded, and monotone mapping from visual-alignment evidence to an intervention strength. The softplus pseudo-counts ensure smoothness and positivity, the concentration parameter c controls the prior strength, and the clamp range provides safety rails against degenerate gates.

Why reinforce high-alignment states. The gate should be interpreted as a trust estimator for the current visual trajectory, not as an error magnitude estimator. When $s_t = \cos(\mathbf{h}_{l,t}, \mathbf{v}_{\text{CARD}})$ is high, the hidden state is already aligned with the visual evidence extracted from prefill, so adding the CARD direction reinforces grounded generation. When s_t is low, the token may be syntactic or the representation may be unstable; reducing the intervention avoids over-steering and helps preserve fluency and recall.

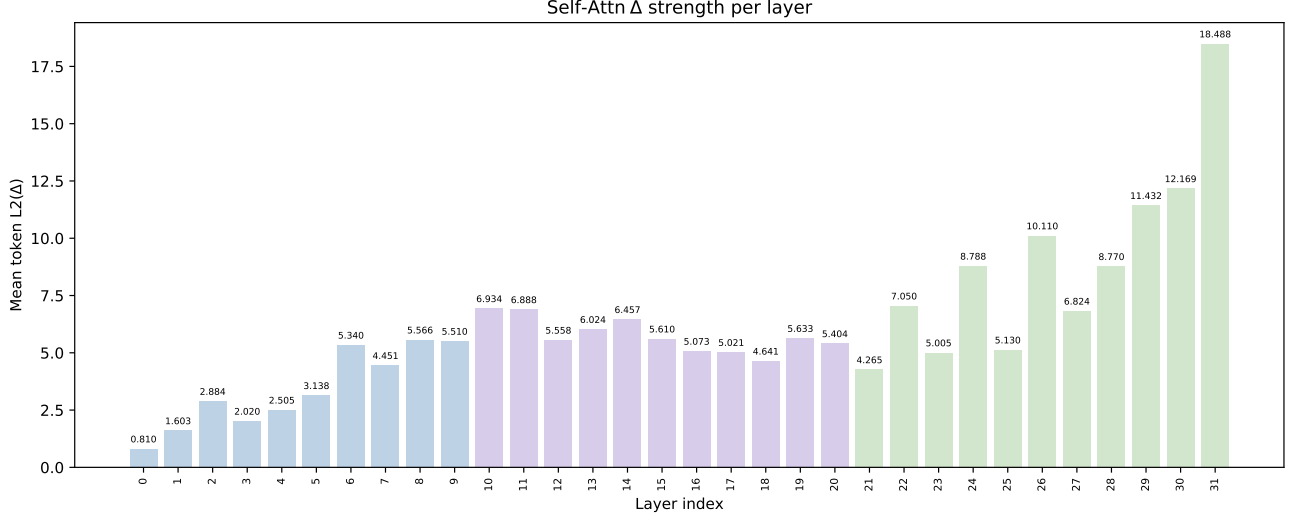
Layer selection across architectures. The best intervention layer is architecture-dependent. For LLaVA-style models and Idefics2, visual tokens are concatenated or projected into the LLM input, and the visual signal tends to fade in middle-to-late language layers; hence late layers offer stronger leverage. For Q-Former-based or stronger early-fusion models such as InstructBLIP and Qwen2.5-VL, useful multimodal alignment can occur earlier, so early intervention can be preferable. We therefore use a small held-out sweep to select a fixed layer per backbone and then keep that layer unchanged across benchmarks.

Efficiency claim. “No extra forward pass” means that RUDDER does not invoke an additional model forward beyond the vanilla prefill and autoregressive decoding calls. CARD is extracted from the mandatory prefill pass through a read-only hook, while decoding only adds cheap vector operations and a residual-stream update. This is why RUDDER keeps the end-to-end latency overhead below 4% in our measurements.

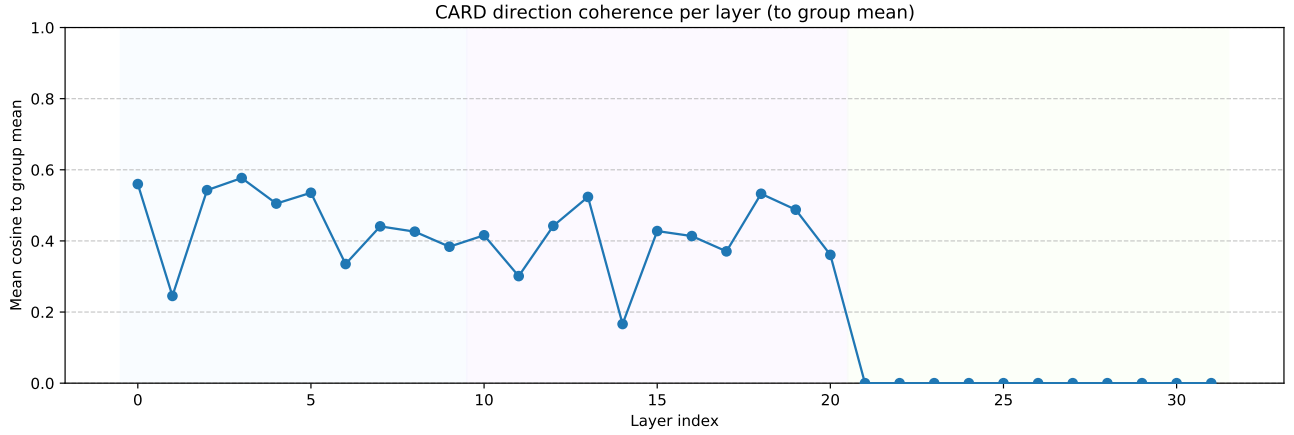
Hallucination–recall trade-off. A method can trivially reduce hallucination by becoming overly conservative. To avoid this failure mode, our calibration explicitly requires at least 95% of baseline recall, and we additionally evaluate general multimodal capability on MME. This ensures that RUDDER reduces unsupported object mentions while largely preserving correct object recall and general visual reasoning behavior.

D. LLMs Usage Statement

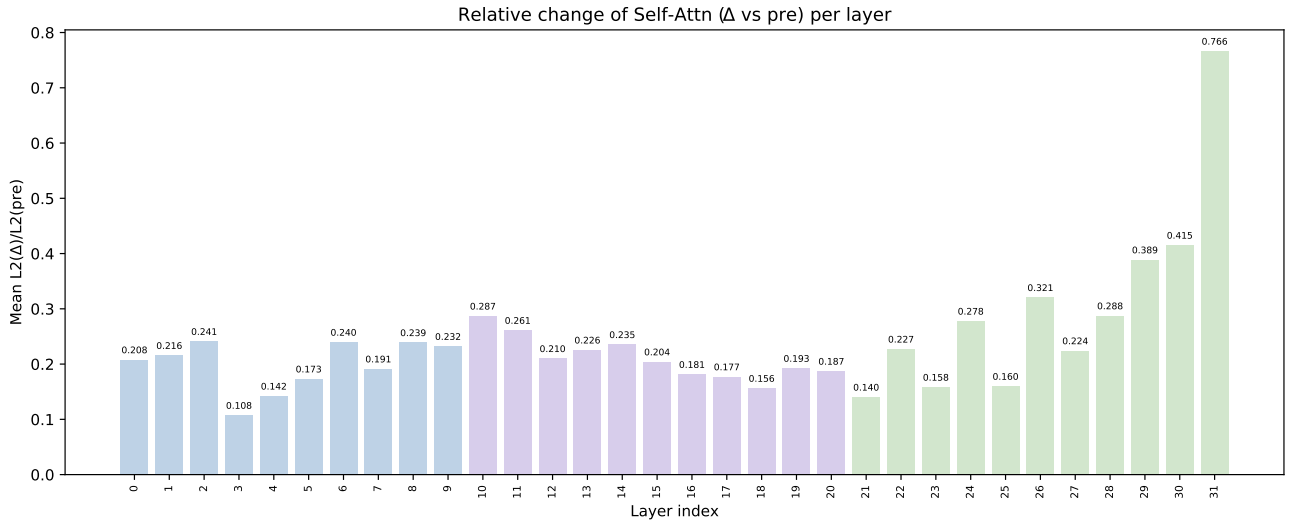
Generative AI has been utilized to enhance the writing and to assist with coding tasks.



(a) Absolute strength of the update vector ($\|\delta^l\|$) across layers.



(b) Directional coherence within layer groups.



(c) Relative strength of the update vector ($\|\delta^l\|/\|h_{\text{pre}}^l\|$).

Figure 9. Analysis of the internal dynamics of LLaVA-1.5 (Liu et al., 2024a). (a, c) Both the absolute and relative strength of self-attention updates peak in the middle-to-late layers, which identify a “computational core”. (b) Mid-late layers show significantly lower directional coherence than other regions.



(a) Input image.

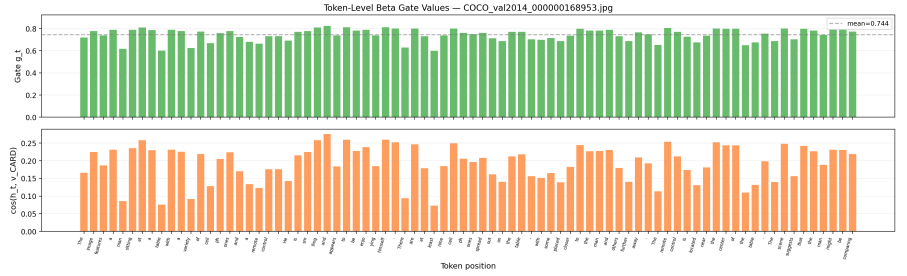


(b) Token-level gate and CARD similarity.

Figure 10. Token-level Beta Gate visualization for a skiing image. Visually grounded tokens receive consistently high gate values, while less visually grounded or more abstract tokens show lower CARD similarity. This supports the interpretation of the gate as a visual-grounding-dependent trust signal.



(a) Input image.



(b) Token-level gate and CARD similarity.

Figure 11. Token-level Beta Gate visualization for a cell-phone image. The hallucinated token sequence related to “remote control” receives low CARD similarity, showing that the gate can down-weight intervention when the current hidden state is poorly aligned with the image-conditioned CARD direction.

Baseline:

"The image features a small, clean kitchen with a white countertop and a stove top oven. The kitchen is equipped with a sink and a microwave, and **there is a toilet visible in the background. The room also has a dining table and a few chairs placed around it.** In addition to the main kitchen appliances, there are a few other items in the room."



RUDDER:

"The image features a clean and tidy kitchen with a variety of appliances and utensils. **There is a sink positioned towards the center of the room, with a microwave oven placed above it.** A refrigerator can be seen on the left side of the room, **while an oven is located towards the bottom left corner.**

Baseline:

"The image features a wooden table with a variety of kitchen utensils and items. **There are several knives, including a large knife and a smaller one, placed on the table.** A spoon is also visible, along with a couple of spoons and a fork. In addition to the utensils, there are multiple bottles on the table, with one being larger and positioned towards the left side, and the others being smaller and scattered around the table. A cup can be seen on the right side of the table, **and a bowl is placed in the middle. table.**"



RUDDER:

"The image features a tabletop with a variety of items laid out on it. **There are several spices neatly arranged in a line.** Additionally, a number of spoons are positioned around the tabletop, some pointing upwards and others pointing downwards. There are also **several bottles dispersed across the tabletop, adding to the assortment of items present.** The scene gives off an impression of a well-organized and functional space for preparing meals or engaging in culinary activities."

Baseline:

"The image features a cozy living room with a couch and a chair. On the couch, there are two cats lying down, one on the left side and the other on the right side. They appear to be relaxed and enjoying their time together. **In addition to the cats, there are two books placed on the couch, one near the left cat and the other near the right cat.** The scene gives off a warm and comfortable atmosphere, with the cats and books adding to the overall ambiance of the living room."



RUDDER:

"The image features two cats lying down on a couch, enjoying each other's company. One cat is positioned towards the left side of the couch, while the other cat is laying down further to the right side. **They both seem to be resting comfortably on newspapers that are spread out on the couch.** The scene creates a cozy atmosphere with the cats sharing their space on the couch."

Baseline:

"The image features a red fire hydrant situated on a sidewalk next to a building. The fire hydrant is prominently placed in the scene, and it appears to be the main focus of the image. The sidewalk is located near a street, and **there is a truck parked on the street, slightly to the left of the fire hydrant.**"



RUDDER:

"The image features a red fire hydrant prominently positioned on the sidewalk near a building wall. The fire hydrant is sitting on top of a sidewalk curbing, making it easily accessible for emergency services. **In addition to the fire hydrant, there are several graffiti tags on the building wall behind it.** These tags add an urban touch to the scene and provide contrasting visual interest against the red fire hydrant."

Figure 12. Case study. Hallucinated contents generated by the vanilla LLaVA-1.5 are marked in red, while the factual contents from our method are colored with blue.

Baseline:

"The image features a man wearing a jacket and a cap, standing on a snowy hill. He appears to be posing for a photo while holding his snowboard. The man **has his hands close to his chest**. The snowy terrain surrounds the man, and in the background, there are additional trees scattered across the area. **A second person is visible in the background, standing further to the right.**"

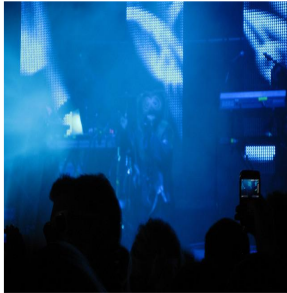


RUDDER:

"The image features a man posing on top of a snow-covered ski slope during winter. He is wearing snow gear and is **standing confidently on skis**. **The scene appears to be captured during the winter season, with the mountainous setting serving as a striking background.**"

Baseline:

"The image captures a lively concert scene with a crowd of people gathered around a stage. A man is standing on the stage, singing into a microphone, **while a woman is also present on the stage, possibly accompanying the singer**. There are several people in the audience, some of them holding cell phones, likely capturing the moment or sharing the experience with others. The atmosphere appears to be energetic and engaging, with the performers and the audience enjoying the concert together."



RUDDER:

"The image features a live concert event **taking place indoors**. At the center of attention is a person singing into a microphone **while a band plays instruments behind them**. There are several other people watching the performance from various positions around the room. Some of these individuals are standing close to each other, while others are spread out across the venue. In addition to the live performance setup, there are multiple cell phones being used by concertgoers to capture memories of the event."

Baseline:

"The image features a busy city street with several police officers patrolling the area on their motorcycles. There are **three police officers** prominently visible in the scene, with one officer positioned slightly to the left of the center, another officer slightly to the right of the center, and **the third officer slightly to the right of the rightmost officer**. Multiple cars are parked or driving down the street, including one car parked on the left side of the image, another car parked further down the street on the right side, and several cars driving down the roadway. Additionally, there are two handbags visible in the scene **one near the leftmost officer and the other slightly to the right of the center officer.**"



RUDDER:

"The image depicts a busy city street filled with various vehicles and pedestrians. **There are two police officers on motorcycles, one on the left side and the other on the right side of the street. They appear to be directing traffic or monitoring the area.** In addition to the motorcycles, there are several cars and a truck on the street. Some cars are parked, while others seem to be in motion. A few pedestrians can be seen walking along the sidewalk, **with one person carrying a handbag**. Overall, the scene captures the hustle and bustle of city life, with the police officers playing a crucial role in maintaining order and safety."

Baseline:

"The image features a black cat sitting in a bathroom sink, surrounded by various toiletries. **The sink is filled with toothbrushes**, some of which are placed near the cat, while others are scattered around the sink. There are also multiple bottles in the scene, with some located near the cat and others placed further away. The cat appears to be relaxed and comfortable in the sink, surrounded by the bathroom items."



RUDDER:

"The image features a black cat comfortably lying down inside a bathroom sink. The sink occupies most of the scene, **with the cat occupying its central space**. The cat appears to be enjoying its time in the sink, possibly finding it cozy or cooler than its surroundings."

Figure 13. Case study. Hallucinated contents generated by the vanilla Idefics2 are marked in red, while the factual contents from our method are colored with blue.

Baseline:

" There are four plates with different types of cakes and pastries, each with a unique flavor and design. The desserts are arranged in a way that they are easily accessible for the diners. **The table is set with utensils, including a fork and a knife,** and a cup is also present. The dining area is well-lit, providing a pleasant atmosphere for enjoying the delicious treats."



RUDDER:

" There are four plates on the table, **each containing a different dessert.** The dessert selection includes a slice of cake, a slice of cheesecake, a slice of lemon cake, and a slice of chocolate cake. The dessert plates are arranged in a way that allows for a variety of flavors to be enjoyed."

Baseline:

" **Two men and a woman present,** all wearing different clothing. One man is holding a kite, **while the other man is holding a bottle.** The woman is standing nearby, possibly assisting with the kite or enjoying the activity. "



RUDDER:

" **There are two kites in the air, one of a kite shaped like a killer whale.** The people are having fun and flying the kites, possibly enjoying the day together."

Baseline:

" He is looking upwards, possibly in a contemplative or thoughtful manner. The room appears to be a dining area or a gathering place, **with a table and chairs visible in the background.** The man's facial expression and body posture suggest a sense of seriousness or concentration "



RUDDER:

"He is looking upwards, possibly at a person or object in the room. The man is the focal point of the scene, and his presence is quite prominent. **There are two lights in the room, one on the left side and another on the right side.** The man is the center of the scene, and his presence is quite prominent."

Baseline:

" The baby is wearing a white outfit, and the man is smiling at the camera. The baby is positioned close to the man's face, making it a heartwarming moment captured in the photo. **There are several other people in the background, including a man and a woman, both standing in the same room. The man and woman are positioned on opposite sides of the room, with the man holding the baby and the woman standing in the background "**



RUDDER:

" The baby is wearing a white outfit, and the man is smiling at the camera. There are two other people in the room, **one on the left side and another on the right side. The room appears to be a church, with a cross visible in the background.** The man and the baby are the focal point of the scene."

Figure 14. Case study. Hallucinated contents generated by the vanilla InstructBLIP are marked in red, while the factual contents from our method are colored with blue.

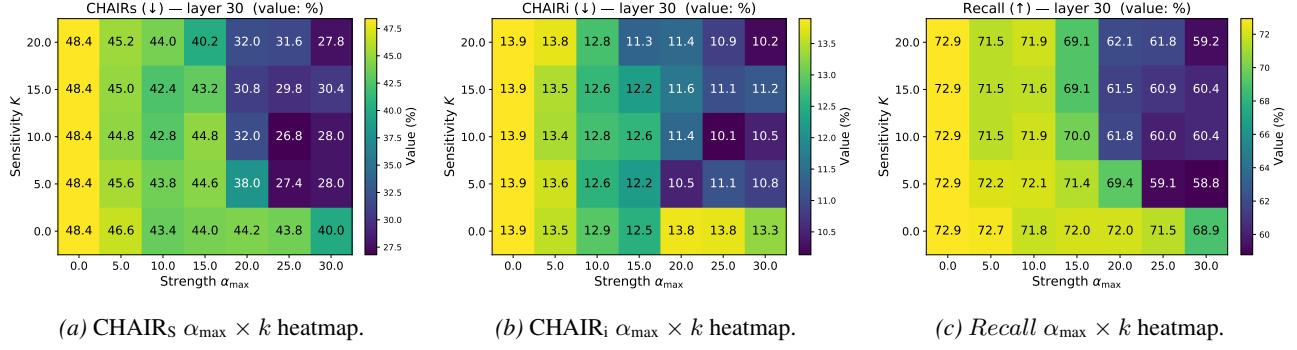


Figure 15. Ablation matrices for RUDDER on LLaVA-1.5 (Liu et al., 2024a)

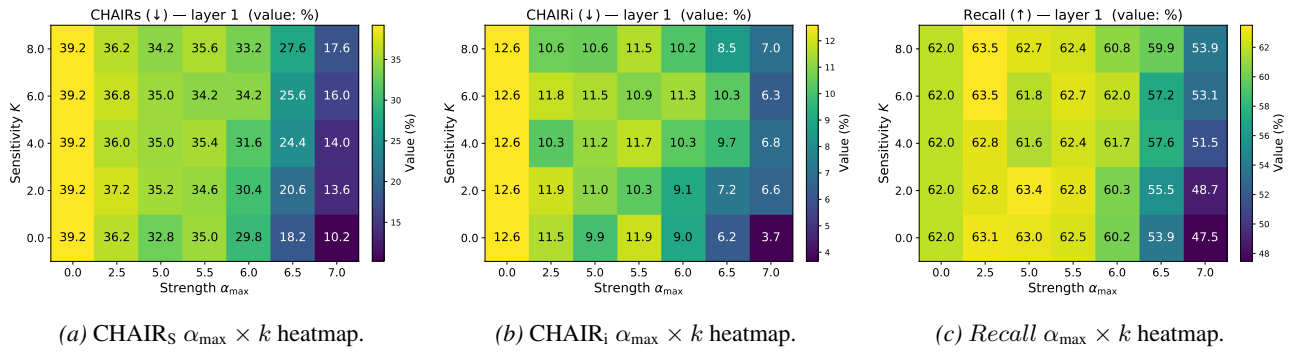


Figure 16. Ablation matrices for RUDDER on InstructBLIP (Dai et al., 2023)